

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-ab7c5e07ea26ec571.jsonl`  
- SHA-256: `9a047bfff98cf6c0f8760258e86e3e04a44019f8c098800e2acbb729a82653480`  
- Source modified: `2026-06-16T03:21:00+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:20:03.177Z)

Source Extractor

Research question: "Research question: **What concrete, recent (2022?2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best *recent* methods for (a) turning a dense per-frame signal into well-segmented actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMM/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k,

edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6716 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6716 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 375 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible."

Fetch and extract key claims from this source:

****URL:**** <https://arxiv.org/pdf/2012.08668>

****Title:**** Mitigating Bias in Calibration Error Estimation (Roelofs et al., AISTATS 2022)

****Found via:**** Small-n evaluation rigor for video segmentation search

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2. assistant (2026-06-16T03:20:05.709Z)

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3. user (2026-06-16T03:20:19.327Z)

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"calibration_fig3.pdf")

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To properly answer your questions about:

1. Bias in standard ECE estimators
2. Recommended alternative estimators
3. Guidance on bin selection and debiasing
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[Binary content (application/pdf, 1.7MB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\tool-results\webfetch-1781580013109-x8y2q3.pdf]

4. assistant (2026-06-16T03:20:22.071Z)

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5. user (2026-06-16T03:20:25.275Z)

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6. user (2026-06-16T03:20:29.648Z)

Analysis of Calibration Error Estimation Paper

Abstract & Key Claims

****Main Finding on Bias:****

The paper identifies that "binning-based estimators with bins of equal mass (number of instances) have lower bias than estimators with bins of equal width."

****Recommended Estimators:****

Two reliable approaches are proposed:

1. The debiased estimator (existing method from Brocker 2012; Ferro & Fricker 2012)
2. ECE_sweep (novel method), which "uses equal-mass bins and chooses the number of bins to be as large as possible while preserving monotonicity"

****Specific Claims About Bins & Debiasing:****

The abstract states these estimators yield "improvements in the effectiveness of recalibration methods and in the detection of model miscalibration," though specific numerical reductions are not quantified in the abstract itself.

****Methodology:****

The framework "involves synthesizing model outputs that have the same statistics as common neural architectures on popular data sets" to assess bias computationally.

****Publication Details:****

- ****Venue:**** AISTATS 2022
- ****Submission Date:**** December 15, 2020
- ****Final Version:**** February 11, 2022 (v3)

- **arXiv ID:** 2012.08668

Code Availability: Implementation available at the referenced GitHub repository (caltrain).

7. assistant (2026-06-16T03:20:33.538Z)

The arXiv abstract page fetch succeeded and gave me the key claims. Let me try to extract more detail from the PDF text body to verify specific quotes and numbers, using a Python-based text extractor.

8. user (2026-06-16T03:20:37.208Z)

Mitigating Bias in Calibration Error Estimation
Rebecca Roelofs Nicholas Cain Jonathon Shlens Michael C. Mozer
Google Research Google Research Google Research Google Research
Abstract

For an AI system to be reliable, the confidence it expresses in its decisions must match its accuracy. To assess the degree of match, examples are typically binned by confidence and the per-bin mean confidence and accuracy are compared. Most research in calibration focuses on techniques to reduce this empirical measure of calibration error, ECEbin. We instead focus on assessing statistical bias in this empirical measure, and we identify better estimators. We propose a framework through which we can compute the bias of a particular estimator for an evaluation data set of a given size. The framework involves synthesizing model outputs that have the same statistics as common neural architectures on popular data sets. We find that binning-based estimators with bins of equal mass (number of instances) have lower bias than estimators with bins of equal width. Our results indicate two reliable calibration-error estimators: the debiased estimator (Brocker, 2012; Ferro and Fricker, 2012) and a method we propose, ECESweep, which uses equal-mass bins and chooses the number of bins to be as large as possible while preserving monotonicity in the calibration function. With these estimators, we observe improvements in the effectiveness of recalibration methods and in the detection of model miscalibration.

1 INTRODUCTION

Machine learning models are increasingly deployed in high-stakes settings like self-driving cars (Caesar et al., 2020; Geiger et al., 2013; Sun et al., 2020) and medical diagnosis (Esteva et al., 2017, 2019; Gulshan et al., 2016) where it is critical to recognize when a model is likely to be incorrect. Unfortunately, models often fail Proceedings of the 25th International Conference on Artificial Intelligence and Statistics (AISTATS) 2022, Valencia, Spain. PMLR: Volume 151. Copyright 2022 by the author(s).

in unexpected and poorly understood ways, hindering our ability to interpret and trust such systems (Azulay and Weiss, 2018; Biggio and Roli, 2018; Hendrycks and Dietterich, 2019; Recht et al., 2019; Szegedy et al., 2013). To address these issues, calibration is used to ensure that a model produces confidence scores that reflect its ground truth likelihood of being correct (Platt et al., 1999; Zadrozny and Elkan, 2001, 2002).

To obtain an estimate of the calibration error, or ECE, the standard procedure partitions the model confidence scores into bins and compares the model's predicted accuracy to its empirical accuracy within each bin (Guo et al., 2017; Naeini et al., 2015). We refer to this specific metric as ECEbin. Recent work observed that the calculation of ECEbin is sensitive to implementation (Kumar et al., 2019; Nixon et al., 2019). Fundamentally, a key confounding factor is statistical bias, the difference between the expected ECEbin and the true calibration error (TCE). Because bias is largely unexplored in the literature, its magnitude and sign is unknown, as is its dependence on hyperparameters of the ECEbin estimator (e.g., number of bins, how bins are formed). We explain our reasons for focusing on estimator bias and not variance in Section 4.

Bias in ECEbin measurement has two real world consequences. First, the measurement of calibration error on a given model may be systematically incorrect. Thus, our understanding of how well a model knows whether it is correct may be poor, and may not be accurately captured by naively reporting ECEbin. Second, many techniques have been developed to minimize the calibration error, such as post-hoc recalibration techniques (Guo et al., 2017; Zadrozny and Elkan, 2001, 2002) and, more recently, calibration-sensitive training objectives (Karandikar et al., 2021; Krishnan and Tickoo, 2020; Kumar et al., 2018; Lin et al., 2018; Mukhoti et al., 2020). Given that the selection of the training objectives and the justification of a recalibration technique is predicated on the measurement of the calibration error, Naeini et al. (2015) introduce ECE as an acronym for Expected Calibration Error. However, ECE is not a proper expectation whereas the true calibration error is computed under an expectation. To resolve this confusion, we prefer to read ECE as Estimated Calibration Error.

arXiv:2012.08668v3 [cs.LG] 11 Feb 2022 Mitigating Bias in Calibration Error Estimation

0.0 0.2 0.4 0.6 0.8 1.0

Predicted confidence

0.0

0.2

0.4

0.6

0.8

1.0 Empirical accuracy

Sample B

Raw data

True calibration

ECE Bins

0 5 10 15 20

ECEBIN (%)

0.00

0.05

0.10

0.15

0.20 PDF

ECEBIN Distribution

Mean

Sample B

(a) Figure 1: ECEbin exhibits large bias for perfectly calibrated models. We simulate data from a perfectly calibrated model with confidence scores θ to ResNet-110 CIFAR-10 output (He et al., 2016; Kängsepp, 2019) and measure ECEbin using 15 equal-width-spaced bins. The left panel shows a reliability diagram for a sample of size $n = 200$ (named Sample

B); the right panel has a distribution of ECEbin scores computed across 106 independent simulations. Even though the model is perfectly calibrated, ECEbin systematically predicts large calibration errors. error, reliance on an inaccurate estimator may lead to a suboptimal choice.

We address this problem by developing a technique to measure bias in calibration metrics, which we call the bias-by-construction (BBC) framework. The BBC framework uses simulation to create a setting where the TCE can be computed analytically and thus the bias can be estimated directly. BBC reveals that ECEbin has systematic non-negligible statistical bias, particularly for perfectly calibrated models (Figure 1).

Our goal is to identify the least biased estimator of calibration error using BBC. We consider two estimators previously proposed in the literature: the debiased estimator (Brocker, 2012; Ferro and Fricker, 2012), which we refer to as ECEdebias, and the smoothed kernel density

9. assistant (2026-06-16T03:20:39.992Z)

I now have the verbatim abstract and introduction text with strong, falsifiable claims. I have enough to extract the claims with direct quotes.

10. user (2026-06-16T03:20:50.499Z)

Structured output provided successfully

11. assistant (2026-06-16T03:20:59.928Z)

I've extracted and verified the claims from the source. The fetch succeeded (verbatim abstract plus introduction text confirmed directly from the PDF body), so this is a primary research source.

Summary: Roelofs et al. (AISTATS 2022), "Mitigating Bias in Calibration Error Estimation," is directly relevant to Dimension 4 (evaluation rigor at small n). Key takeaways for the badminton rally-detector leaderboard:

- Plain binned ECE has systematic, non-negligible bias that over-reports calibration error, worst for well-calibrated models ? so reporting raw ECE on ~6716 videos is untrustworthy.
- Use equal-mass bins (not the default equal-width) for lower bias.
- Adopt one of two reliable estimators: the debiased estimator (Brocker 2012; Ferro & Fricker 2012) or their ECE_sweep (equal-mass bins, max bins preserving monotonicity).
- The bias is acute at small evaluation sizes (their demo uses n=200), reinforcing that the current Platt/isotonic + Brier/ECE harness should switch off naive equal-width ECE.

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